

ASHISH PATIL

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SUMMARY

Full-Stack AI Engineer passionate about building intelligent systems, developer tools, and real-world products. Interested in building AI-powered systems, large language models, backend engineering, and scalable software architecture, while combining strong engineering principles with creative UI/UX design to deliver intuitive user experiences. Enjoy building products from idea to production by blending engineering, design, and practical AI into software that solves real-world problems.

EDUCATION

Pimpri Chinchwad College of Engineering, Pune

B.Tech in Computer Engineering

Aug 2023 – Present

CGPA: 7.84 / 10

Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database & Management Systems, Machine Learning, Deep Learning, Generative AI, Cloud Computing

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript, SQL

AI / ML: LangChain, LangGraph, LangSmith, RAG Pipelines, Gemini API, OpenAI API, Agentic Systems, LLM Security

Frontend: React.js, Next.js, Tailwind CSS, REST APIs, WebSocket

Backend: Node.js, Express.js, FastAPI, Pydantic, Event-Driven Architecture

Databases: PostgreSQL, MySQL, MongoDB, Firebase, Supabase

Cloud & DevOps: AWS (EC2, S3, RDS), Google Cloud, Git, Linux, CI/CD, GitHub Actions

PROJECTS

Scankii: AI Agent Security Scanner

Python · PyPI · GitHub Actions

- Built the first open-source security scanner for AI agent skills that jointly analyzes source code and natural-language instructions to detect cross-modal credential leaks, prompt injection, MCP supply-chain attacks, and other AI-specific vulnerabilities before deployment.
- Achieved **100% precision, 68.2% recall, F1 81.1%** on SkillLeakBench (Chen et al., ASE 2026) without dynamic sandbox execution; ships `scankii.runtime` for auto-redacting credentials before they reach the LLM context window, a GitHub Action with SARIF upload, and a pre-commit hook.

Guardian-Runtime: AI Agent Security Proxy

Python · FastAPI · Pydantic · PyPI

3,000+ PyPI downloads · 18 GitHub Stars · Published on PyPI

- Built a local security proxy and firewall for autonomous AI coding agents, intercepting every LLM prompt and response at the transport layer to enforce FinOps budgets, prevent credential leaks, and compress token payloads entirely on-device before any data leaves the machine.
- Architected a two-layer token optimization engine achieving **40–70% reduction in output token costs**; built pluggable FastAPI/Pydantic middleware supporting OpenAI, Gemini, and Claude with configurable per-session budget limits and zero-code provider switching.

CET Buddy: College Predictor Platform

JavaScript · Python · Data Engineering

- Engineered a fully client-side college prediction engine parsing **2,20,000+ MHT-CET cutoff records (2023–2026)** in-browser with sub-second filter response across percentile, category, branch, and college dimensions, with paginated results and one-click PDF export.
- Built a Python data pipeline normalizing government-published datasets across 3 academic years; scaled to 1,000+ users on a mobile-first UI with **~80% reduction in manual research time** for Maharashtra students, with zero server costs and zero paid acquisition.

KodeBattle: Real-Time Competitive DSA Platform

React · Node.js · WebSocket · Firebase

- Architected a real-time 1v1 competitive programming platform with **event-driven game state synchronization** over persistent WebSocket connections; deterministic result delivery across Ranked, Private Duel, and AI Bot matchmaking modes.
- Implemented an Elo-style KOD rating system with persistent match history, real-time analytics (rating graphs, activity heatmaps via Firestore), and an anti-cheat layer with fullscreen enforcement and 3-strike automated forfeit logic.

ACHIEVEMENTS & COMPETITIVE PROGRAMMING

- Open Source:** Merged contributor to the **Awesome Vibe Coding Resources** repository (**235+ GitHub stars**).
- Community Engagement:** Participated in technical design discussions with the **Microsoft AutoGen open-source community**, incorporating external architectural feedback and independently validated improvements into **Scankii**.
- Coding Profiles:** **LeetCode** Contest Rating **1627** | **CodeChef** 3★ Rating **1635** (Div. 2)
- Research:** Research paper on AI-powered academic certificate verification integrating OCR, deep learning, and blockchain for document authentication and forgery detection.
- Finalist, Top 50:** at Amalthea National Hackathon (IIT Gandhinagar × OdoO), selected from 200+ teams nationwide.